

Forecasting solar cycle 25 using comprehensive precursor combination and multiple regression technique

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ABSTRACT

In this paper, we propose a new model to predict the complete sunspot cycle based on the comprehensive precursor information (peak, skewness, maximum geomagnetic index aa of the previous cycle, and start value of predicted cycle). The monthly average sunspot original data are processed by Gaussian smoothing and the new model is validated by the observed sunspots of cycle 24. Compared with the traditional 13-month moving average, the Gaussian filter has less missing information and is better to describe the overall trend of the raw data. Through the permutation and combination of multiple parameters in precursor methods of solar cycle forecasting, the multiple regression technique is used to successfully achieve the peak prediction. The regression coefficient (R) of the empirical model established in this paper can reach 0.95. By adding a new parameter to the original HWR function, we provide a complete solar cycle profile showing unimodal structure. It shows that the peak value of cycle 25 will come in March 2024, with a peak of 140.2.

Key words: Sun: activity – sunspots.

1 INTRODUCTION

As we all know, sunspots that can represent solar activity have a long history of research. There are many researches on sunspot variation and solar cycle prediction. The prediction of the sunspot cycle can be classified as following three types: The first is the time series method. In this method, the sunspot number is regarded as a function of time for statistical prediction, in which the observed value or smoothed value of the sunspot number is regarded as a non-stationary random time series (Solanki 2003). The traditional approach is to transform the non-stationary into stationary time series, such as the famous McNish–Lincoln method (McNish & Lincoln 1949) and Box–Jenkins method (Box & Jenkins 1990). These kinds of the method have not yet found an ideal way to accomplish the transformation. It can only get the general trend of solar activity with the help of the periodicity of the series itself but cannot get satisfying prediction accuracy. The neural network method developed in the last decade is also used in forecasting. For example, Chattopadhyay et al. (2011) found that the effect of the auto-regressive neural network (ARNN) model in forecasting the solar cycle is better than the traditional time series method. With the sunspot number from 1749 to 2019, Pala & Atici (2019) used a long-term memory algorithm and neural network auto-regressive (NNAR) algorithm to predict the solar activity in the 25th cycle, showing that sunspots will reach the peak between 2022 and 2023. Although the neural network method sometimes performs better than many classical time series methods, however usually it is just like a black box without physical meaning and the performance also needs further validated. Similar to the above method is the time

curve fitting method, Wilson (1988, 1992) proposed the concept of long and short sunspot cycles, by using three sinusoidal curves to fit the sunspot number curve and extrapolate the predicted value. Some people also used other curves or multiple line regression methods to predict the sunspot number (Nordemann 1992; Hathaway, Wilson & Reichmann 1994). In particular, Hathaway et al. (1994) proposed a simple function (HWR function) to approximate the solar cycle profile with a certain predictive function. This kind of method aims at the profile prediction of a solar cycle, which is suitable for the prediction of the solar activity in the declining phase of the cycle, but the prediction for the peak value of the cycle is poor and not robust.

The second is the activity cycle parameter method. This method is based on the empirical relationship between the parameters of different active cycles or active cycles, such as the extremum, the time length of the rising or falling period (Hathaway, Wilson & Reichmann 2002; Vaquero, García & Gallego 2006). It mainly predicts the maximum of sunspots and its time position in the future solar cycle. Hathaway et al. (2002) found that there is also a positive correlation between the peak and valley of the same cycle called the maximum–minimum effect. Most empirical formulas used in this kind of forecast are obtained by the methods of probability statistics and correlation analysis. Hathaway et al. (2002) introduced the concept of high and low value week, odd and even weeks and use different forecast formulas for different weeks. Vaquero et al. (2006) introduce the concept of fast-rising and slow-rising cycles to forecast.

The third is the precursor method. This method is used to forecast the maximum number of sunspots in the next solar cycle by using the measurement of geophysical quantities in the minimum period or the solar cycle falling period. For example, Lantos & Richard (1998) proposed that the maximum aa index ($aa_{\max}$) in the last 4 yr before the end of the cycle is significantly correlated with the peak value of

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the next cycle, and the geomagnetic index has a strong correlation with the sunspot peak (the regression coefficient is usually above 0.9). This precursor prediction method of geophysical phenomena can be explained physically. It connects the minimum geomagnetic activity with the maximum geomagnetic activity through the minimum solar magnetic field. Another precursor prediction method uses the relationship between the internal properties of the solar cycle, called the precursor method of solar activity phenomenon. This approach is based on the understanding that the solar activity is mainly the magnetic activity, and the strong magnetic activity in the middle and low latitudes must be evolved from the weak magnetic activity in the polar region or high latitudes at the beginning of the solar cycle. Therefore, according to some measurements in the minimum period (such as the latitude of the new cycle sunspot, the intensity of the polar magnetic field, etc.), one should be able to predict the maximum period of sunspot activities. Generally, compared with the other information, the geomagnetic index has the highest statistical correlation with the period peak to be predicted. However, no matter what kind of prior information, it will produce significant errors in the prediction of very few individual periods (20 per cent deviation from the real observation value as the standard). In the 23rd solar cycle, the average value of all the published prediction results based on geomagnetic activity index information was 157. Compared with the real observation result of 120.7, it is obvious that the prediction is a failure. The forecast for cycle 24 is even much worse, especially for the peak occurrence time (Svalgaard, Cliver & Kamide 2005; Dikpati, De Toma & Gilman 2006; Kane 2007; Watari 2008; Ahluwalia & Ygbuhay 2010). It is difficult to get the prediction result that is consistent with the actual observation value by using only one geomagnetic activity index.

For the coming cycle 25, different people gave completely different prediction results (Bhowmik & Nandy 2018; Okoh et al. 2018; Upton & Hathaway 2018; Brun et al. 2019; Labonville, Charbonneau & Lemerle 2019; Pala & Atici 2019). For example, Bhowmik & Nandy (2018) suggested that the cycle maximum comes in 2024 ± 1 while Labonville et al. (2019) believed the peak in 2025, and Pala & Atici (2019) show between 2022 and 2023. Based on the monthly average sunspot data from 1867 February to 2020 October, this paper combines the prediction of the peak value of the solar activity cycle based on the comprehensive prior information with the prediction of the outline of the solar activity cycle based on HWR, analyses the correlation among the parameters in the cycle, and makes a statistical prediction of the 25th cycle of sunspots. The paper is organized as follows. In Section 2, we introduce the data used in this research. Section 3 introduces the prediction method used in our research. Section 4 is the verification of the method. Section 5 presents the result of our study. A brief summary and discussion are given in Section 6.

2 DATA

The sunspot number data comes from the website <https://omniweb.gsfc.nasa.gov/>. Since the sunspot number data before 1850 are artificially reconstructed rather than observed, this research only selects the observed monthly average sunspot number from 1867 February to 2020 October (1844 months in total), in which the data from the 23rd cycle and before, are used for modelling construction, and the data of cycle 24 are used for verification. The geomagnetic activity aa index data, which comes from http://isgi.unistra.fr/data_download.php/, are also used in the modelling.

In order to better understand the stable law of solar cycle change, it is necessary to filter out the high-frequency variation components

by smoothing the actual observed value of sunspot number before modelling. This smoothing process usually uses ‘13-month running mean’ method (Hathaway, Wilson & Reichmann 1999), and the calculation method is shown in equation (1).

$$R_i^{\text{smooth}} = \frac{1}{12} \left(\sum_{j=-5}^5 R_{i-j} + \frac{1}{2} (R_{i-6} + R_{i+6}) \right), \quad (1)$$

where R_i^{smooth} represents the smoothed average number values of the i -th month, R_i is the averaged sunspot number values of the i -th month.

Although the number of sunspots processed by the 13-month running average can weaken the impact of sudden fluctuations on the overall trend, missing the data of the first 6 months and the last 6 months cannot always reflect the trend. To make up for this lack, we use Gaussian filter to smooth the monthly average sunspot number. The value of each pixel is obtained by weighted average of its own and other pixel values in the neighbourhood, as shown in the following:

$$G(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}, \quad (2)$$

where x indicates the coordinate point in the one-dimensional space and σ represents the Gaussian filter scale (we take $\sigma = 3.6$). Then the value of each position on the mask and the number of sunspots in the corresponding position on the image do multiply-add operations (Davies 1990).

Fig. 1 shows the original monthly mean value of the sunspot number from 1867 February to 2020 October (yellow curve) and the smoothed monthly mean values provided by the two smoothing methods. Compared with the monthly mean value sequence curve of sunspots, the smoothed monthly mean curves by the two methods effectively reduce noise while maintaining the original data change trend, and the Gaussian smoothing monthly mean curve, in which there is less missing data, fits the extreme value and the overall trend better. Obviously, the Gaussian smoothing method is able to provide more accurate skewness information, therefore is more suitable for forecasting. For description simplicity, the monthly mean value of the sunspot number mentioned below refers to the value processed by Gaussian filter unless otherwise specified.

3 FORECAST METHOD

There have been many previous studies on describing the contour features of the solar cycles based on precursor information. The precursor methods mainly make use of the statistical correlation between the known information and the feature parameters to be estimated, to establish effective linear-regression models, and then to achieve the prediction of feature parameters. Depending on what information is available, precursor methods can be mainly classified as implicit information (Hathaway et al. 2002; Li, Gao & Su 2005; Vaquero et al. 2006), geometric information (Lantos 2000, 2006), and geomagnetic activity index information (Bounar, Cliver & Boriakoff 1997; Lantos & Richard 1998; Mendoza & Ramirez 1999).

Fig. 2 shows the correlation between precursor information and peaks of solar cycles 11 to 23, along with the results of a linear fitting (black line). Figs 2(a) and (b) are the linear regression relationships based on the implicit information as the known information. Fig. 2(a) shows the scatter plot and fitting curve between the odd-numbered periodic peak [$RI_{\text{max}}(n)$, the maximum peak of the n th cycle] and the corresponding even-numbered periodic peak of the previous cycle [$RI_{\text{max}}(n-1)$] based on the Even–Odd effect proposed by Hathaway

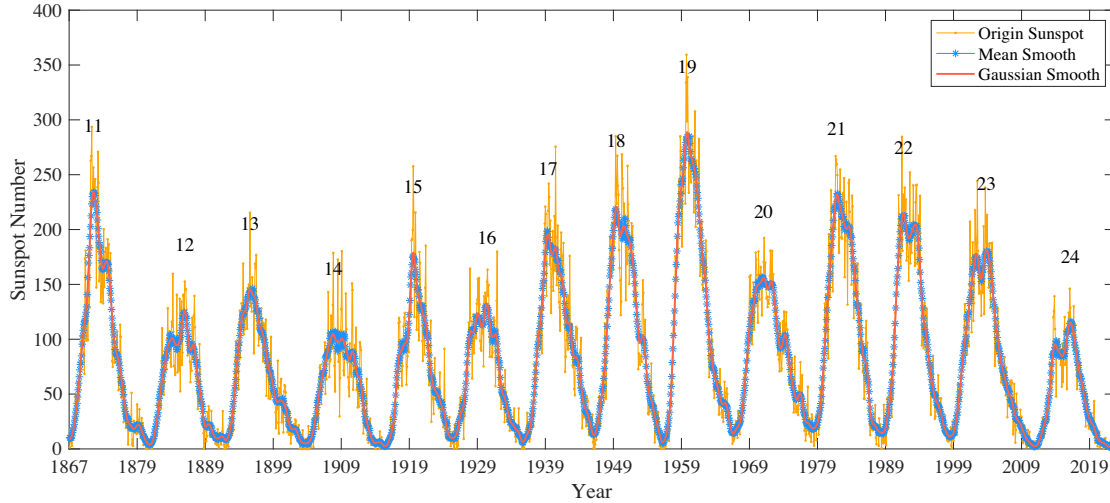


Figure 1. The original monthly mean value of sunspots from 1867 February to 2020 October and their smoothed values. The original monthly mean value of sunspots is depicted by the yellow curve, the smoothed monthly mean value of sunspots after Gaussian filter smoothing is represented by the red curve, and the smooth monthly mean value of sunspots processed by the 13-month moving average method is shown by the blue curve. Take the 24th cycle as an example, the peak period of the original data is in 2014 February, the peak period after Gaussian filtering is 2014 March, and the peak period after 13-month running average processing is in 2014 April.

et al. (2002). As we can see from the figure that $RI_{\max}(n)$ and $RI_{\max}(n-1)$ have a positive correlation (here n is odd) and their correlation coefficient R is 0.61. The Maximum–Minimum effect proposed by Hathaway et al. (2002) indicates that there is a significant correlation between valleys and peaks in the same period. Fig. 2 (b) shows the scatter plot and fitting curve of $RI_{\max}(n)$ and $RI_{\min}(n)$ in the same cycle. As shown in Fig. 2(b), a larger valley value $RI_{\min}(n)$ corresponds to a larger peak value $RI_{\max}(n)$, and the correlation coefficient R is 0.41. The correlation coefficients of linear regression models in Figs 2 (a) and (b) are not high, so it is not good to apply them to predict the peak value of the next solar cycle. In Fig. 2 (c), the geometric feature information of the solar cycle profile is taken as the known information. As Ramaswamy suggested in 1977, the skewness value of the cycle profile, $s(n-1)$, is considered as the independent variable, and the ratio of the current cycle peak to the previous period peak, $RI_{\max}(n)/RI_{\max}(n-1)$, is the dependent variable. It can be seen that they have a significant correlation and the correlation coefficient R is -0.75 , indicating that the skewness of the circle profile of sunspot time series can be used as the precursor information of the next solar cycle peak. In Fig. 2 (d), we build a linear regression relationship based on the level of geomagnetic activity as the known information. According to the conclusion of Lantos & Richard (1998), the current cycle peak value [$RI_{\max}(n)$] is significantly related to the maximum value of the aa index in the last 4 yr of the previous cycle [$aa_{\max}(n-1)$]. We provide the scatter plot and fitting curve of $RI_{\max}(n)$ and $aa_{\max}(n-1)$ in Fig. 2(d). A larger $aa_{\max}(n-1)$ highly corresponds to a larger peak value $RI_{\max}(n)$ and the correlation coefficient is 0.74. It is suggested that the geomagnetic activity level can also be a precursor for the future solar cycle. In comparison with geometric information and geomagnetic activity index information when using linear regression technology for analysis, the implicit information has the weakest linear correlation with the cycle peak (generally below 0.70). Therefore, this type of information can only provide the general trend of peak changes and is auxiliary information in forecasting, but not appropriate to be independently used in peak prediction. The geometric information and the geomagnetic activity index information both have strong correlations with the periodic peak. In particular, the index-type

information of geomagnetic activity can also give a certain physical explanation when using as precursor information. Therefore, the precursor method based on geomagnetic activity index information is a practical and effective method for predicting the peak of solar cycle.

Based on the above analysis and previous experience, this paper selects all relevant parameters [$RI_{\max}(n-1)$, $RI_{\min}(n)$, $s(n-1)$, and $aa_{\max}(n-1)$] as precursor information. We take all kinds of combinations of these parameters as precursor information combinations. Then we use the least-square method to do multiple linear regression between the predicted cycle peak and the precursor information combination. Table 1 shows four combinations of precursor information that are effective in predicting the peak value of the next solar cycle.

The combination 1 model not only passes the F test but also has the largest regression coefficient. So we choose the multiple linear regression model of combination (1) to predict the solar cycle peak. Our prediction model is as follows

$$\begin{aligned}
 RI_{\max}(n) = & 1/[b(0) + b(1)aa_{\max}(n-1)s(n-1) \\
 & + b(2)aa_{\max}(n-1)RI_{\max}(n-1) \\
 & + b(3)RI_{\min}(n)s(n-1) \\
 & + b(4)RI_{\min}(n)RI_{\max}(n-1) \\
 & + b(5)aa_{\max}(n-1)^2], \quad (3)
 \end{aligned}$$

where $b(i)$, $i = 0 \dots 5$ represents the parameters to be calculated, $RI_{\max}(n)$ is the peak value of the sunspot prediction in the n th cycle, $RI_{\min}(n)$ indicates the sunspot valley value in the n th cycle. The valley year of each solar cycle is usually taken as the starting point of a solar cycle, the corresponding sunspot number is the valley value of this cycle. $s(n-1)$ represents the skewness of the sunspot profile in the $n-1$ th cycle, and $aa_{\max}(n-1)$ indicates the maximum value of the aa index in the last 4 yr before the end of the $n-1$ th cycle.

The parameter $b(i)$ in equation (3) can be calculated from the peak value of the known solar cycles and precursor information. Note that there are more than six known cycles but only six parameters to be

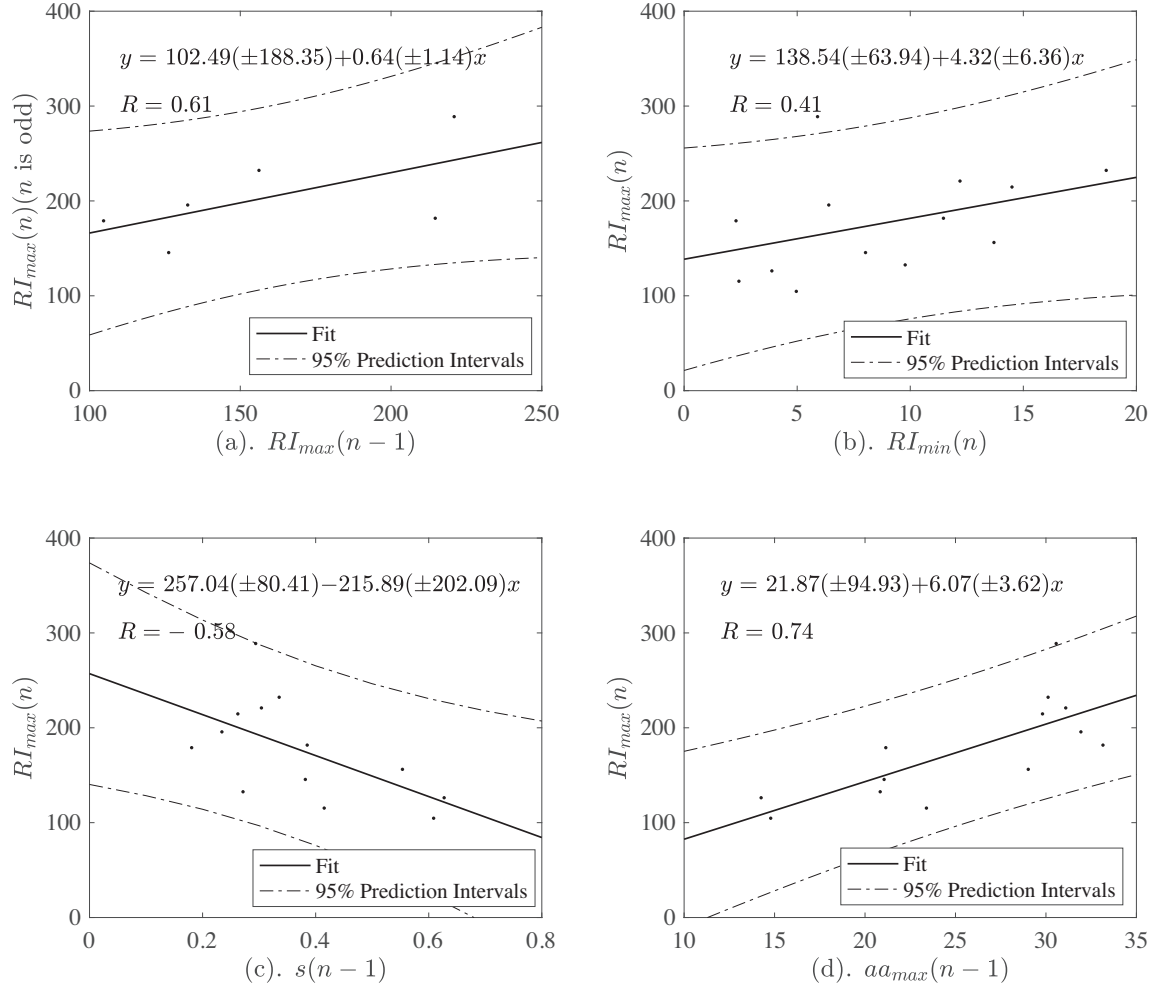


Figure 2. The scatter plot and fitting curve between the precursor information and the peak of the period from the 11th to the 23rd solar cycles. The fitting curve between them is shown by the black curve, the dotted line represents the confidence interval. Figs 2 (a) and (b) show linear regression relationships established based on geometric information [$RI_{\max}(n-1)$, $RI_{\min}(n)$] as the known information, Fig. 2 (c) is a linear regression relationship established based on the geometric information ($s(n-1)$) as the precursor information, Fig. 2 (d) is a linear regression relationship established based on the geomagnetic activity index information [$aa_{\max}(n-1)$] as the precursor information.

Table 1. Multiple linear regression models between four groups of precursor information and cycle peak.

Combination	Precursor information	Regression coefficients	Whether to pass the F test
1	$aa_{\max}(n-1)s(n-1) + aa_{\max}(n-1)RI_{\max}(n-1) + RI_{\min}(n)s(n-1) + RI_{\min}(n)RI_{\max}(n-1) + aa_{\max}(n-1)^2$	0.9505	Yes
2	$aa_{\max}(n-1) + s(n-1) + RI_{\min}(n) + aa_{\max}(n-1)RI_{\min}(n) + RI_{\min}(n)s(n-1)$	0.8738	Yes
3	$aa_{\max}(n-1) + RI_{\min}(n) + RI_{\max}(n-1) + aa_{\max}(n-1)RI_{\min}(n) + s(n-1)^2$	0.8692	Yes
4	$aa_{\max}(n-1)RI_{\min}(n) + RI_{\min}(n)RI_{\max}(n-1) + s(n-1)RI_{\max}(n-1)$	0.8961	Yes

solved, the solution is not unique. We choose the set of solutions with a high regression coefficient and passing the F test. More detail can be found in Section 4. After the parameters to be calculated $b(i)$ and precursor information corresponding to the forecast cycle is known, the peak value of the next cycle can be obtained by equation (3).

After predicting the peak value of the solar cycle, the complete solar cycle can be predicted by the HWR function, one of the most successful methods in function description for studying the periodic changes of solar activity (Hathaway et al. 1994). Since the sunspot valley value is set to zero in the original HWR function and the actual

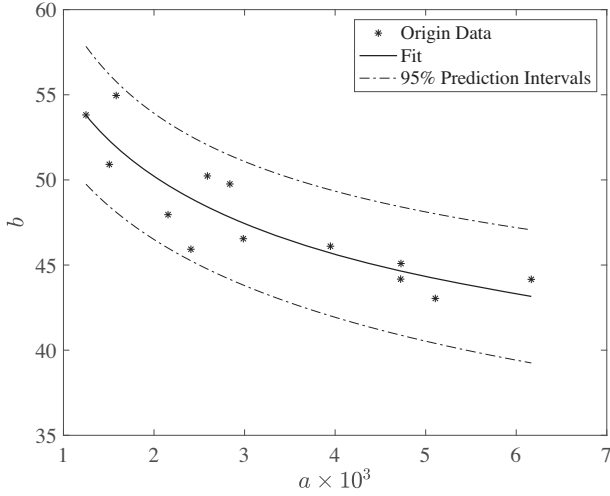
valley is not necessarily at zero, we make a slight improvement to the original model and the complete cycle profile is as

$$f_n(t) = a \frac{(t - t_0)^3}{\exp((t - t_0)^2/b^2) - c} + RI_{\min}, \quad (4)$$

where the parameter a represents the peak value of the cycle, which is directly related to the rising rate of the cycle rising phase, the parameter b is the duration of the cycle rising phase (calculated in months), the parameter c is a quantity which is related to the symmetry of the cycle, and the parameter t_0 represents the beginning time of the cycle.

Table 2. Expected parameter $b(i)$ in cycle 24.

$b(0)$	$b(1)$	$b(2)$	$b(3)$	$b(4)$	$b(5)$
0.006950	0.001099	-0.000002	-0.002012	0.000003	-0.000003

**Figure 3.** Scatter plots of parameters a and b and their fitting curves.

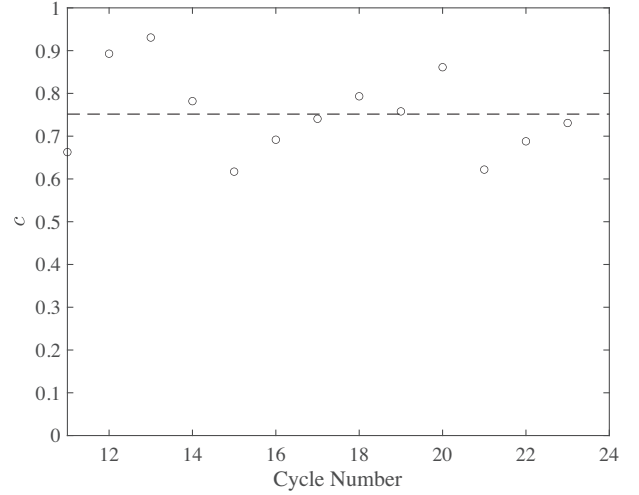
Using the known monthly mean value of sunspots for each cycle, we can fit the parameters a , b , and c through the least mean square error fitting algorithm under nonlinear conditions (Lenvenberg-Marquardt algorithm) and the predicted peak of the solar cycle in equation (3). As a consequence, by combining the predicted peak based on comprehensive precursor information (equation 3) and the improved HWR function (equation 4), the single peak model proposed in this paper can be used to predict the complete sunspot cycle. See Section 4 for the detailed process.

4 MODEL VALIDATION FOR CYCLE 24

For verification of the prediction model, we exploit the monthly mean of sunspots from the cycle 11th to 23rd and their corresponding precursor information (13 sets of data) to predict the 24th solar cycle. Through the prediction method in Section 3, for equation (3), we use the nonlinear least-square method to fit the multiple linear regression model based on the precursor information and the relevant peak value of the 11th and 23rd cycle, obtaining the result of parameter $b(i)$ (see Table 2) as follows: The regression coefficient $R = 0.9415$, the probability that the $F(1, n - 2)$ distribution is greater than the F value, $p = 0.0084 < \alpha$, hence the model is valid. Taking $n = 24$, $aa_{\max}(23) = 23.3942$, $s(23) = 0.4151$, $RI_{\min}(24) = 2.4235$, and $RI_{\max}(23) = 181.7440$, then substitute into the equation (3), we can get that the predicted sunspot peak in cycle 24, $RI_{\max}(24)$ is 121.2740 and the observed sunspot peak is 115.3522, that is the error is less than 6 per cent.

Combined with the improved HWR function, we investigate the solar cycle profile of cycles 11 to 23. For equation (4), the least mean square error fitting algorithm under non-linear conditions, Lenvenberg-Marquardt algorithm, is used to fit the monthly mean value of sunspots in cycles 11 to 23, and the profile model parameters a , b , c of each cycle can be obtained.

Fig. 3 shows the scatter plot and its fitting curve (dotted line) of the profile parameters a and b in each solar cycle from cycles 11 to 23 of sunspots. It can be found that there is a significant

**Figure 4.** Scatter plots of parameters c and its fitting curves.

correlation between the profile parameters a and b . Using the least-square method to approximate the regression expression between profile parameters a and b , we have

$$b = 21.2771 + 6.3779/a^{1/4}, \quad (5)$$

with a regression coefficient $R = 0.8847$. The probability that $F(1, n - 2)$ is greater than F , $p = 5.8841e - 05 < \alpha$ (α is 0.05 by default), that is, the regression model is valid.

Fig. 4 shows the scatter plot of parameter c describing the symmetry of the cycle profile and its fitting curve (dashed line). It can be seen that parameter c is relatively independent of the other two parameters and varies around the mean value of 0.7516; therefore, we take $c = 0.7516$ when applying the HWR function to predict the solar cycle.

For the expression (4) of the HWR function, to find the extreme value, we let $f'(t) = 0$, having

$$1 - ce^{-\left(\frac{t-t_0}{b}\right)^2} - \frac{2}{3} \left(\frac{t-t_0}{b}\right)^2 = 0. \quad (6)$$

Take $c = 0.7516$ and use Newton method to solve equation (6), we have

$$\frac{t - t_0}{b} = 1.0673. \quad (7)$$

That is,

$$t_{\max} - t_0 = 1.0673b, \quad (8)$$

where $t_{\max}$ represents the peak time of the predicted solar cycle and t_0 is the beginning time of the predicted cycle.

Substitute equation (8) into (4) to get

$$RI_{\max} = 0.5125ab^3 + RI_{\min}(24). \quad (9)$$

Then, substitute equation (5) into (9) and $RI_{\min}(24) = 2.4235$, we get the equation to predict the cycle profile by using the cycle peak

$$RI_{\max} = 0.5125a(21.2771 + 6.3779/a^{1/4})^3 + 2.4235. \quad (10)$$

From the predicted value $RI_{\max}(24) = 121.2740$ we get above, we can obtain $a = 0.0015$ by solving equation (10). Then from (5), we get $b = 53.6972$. Therefore, the predicted rising period of cycle 24 is

$$(t_{\max} - t_0)/12 = 4.7759 \text{ (yr)}.$$

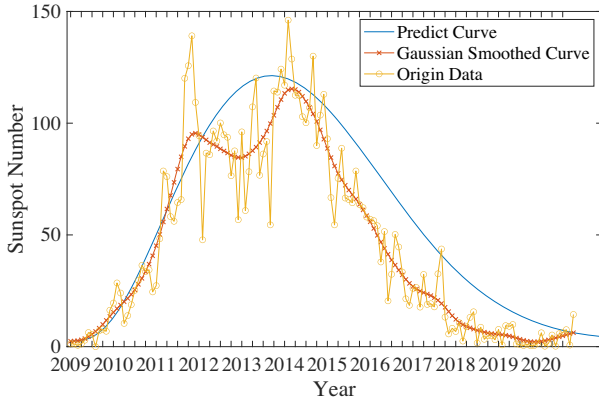


Figure 5. The time-sequence diagram of the 24th cycle of the sunspot and its prediction results. The original monthly mean value of the sunspot is represented by the yellow curve, the smoothed monthly mean value of the sunspot after Gaussian filtering is represented by the red curve, and the monthly mean value of sunspots predicted by the unimodal model is represented by the blue curve.

Table 3. Expected parameter $b(i)$ in cycle 25.

$b(0)$	$b(1)$	$b(2)$	$b(3)$	$b(4)$	$b(5)$
0.006854	0.001191	-0.000002	-0.002218	0.000004	-0.000003

Note that the actual rising period is 5.1667 yr, one can find our model error for the rising period of cycle 24 is within 7.56 per cent, which should be a very good forecast.

Fig. 5 presents the observed and predicted time-sequence diagram of the sunspots number in cycle 24. The prediction results show that sunspots in cycle 24 reach the cycle peak in 2013 October with a peak value of 121.274. By comparing the original month average value of sunspots (yellow line), the smoothed month mean value (red line), and its predicted value (blue line), we found that the prediction performance before the peak is better than near or after the peak. A potential reason is that the sunspot cycle 24 profile has a double-peak structure, and the single-peak model cannot fit this structure well. On the other hand, by first predicting the sunspot peak and then applying it to the HWR function, the error superposition makes the prediction results to be less effective after the peak appears.

5 FORECAST FOR CYCLE 25

To predict the 25th solar cycle we select the monthly mean value of sunspots from the 11th to the 24th cycle and the precursor information of the corresponding cycle (14 sets of data). According to the model in Section 3, for equation (3), we make the non-linear least-square fitting to obtain the multiple linear regression model between the precursor information of the 11th to 24th cycle and the peak value of the corresponding cycle, then get the parameter $b(i)$ listed in Table 3. The regression coefficient $R = 0.9505$, probability of $F(1, n - 2)$ distribution greater than F value, $p = 0.0019 < \alpha$, thus the model is valid. Take $n = 25$, $aa_{\max}(24) = 21.5309$, $s(24) = 0.2521$, $RI_{\min}(25) = 2.1271$, $RI_{\max}(24) = 115.3522$, and get into the equation (3) to obtain the predicted peak value of cycle 25: $RI_{\max}(25) = 140.1562$.

Next, we use the improved HWR function equation (4) to investigate the profile of cycles 11–24. For equation (4), the least mean square error fitting algorithm under nonlinear conditions Levenberg–Marquardt algorithm is used to fit the monthly mean

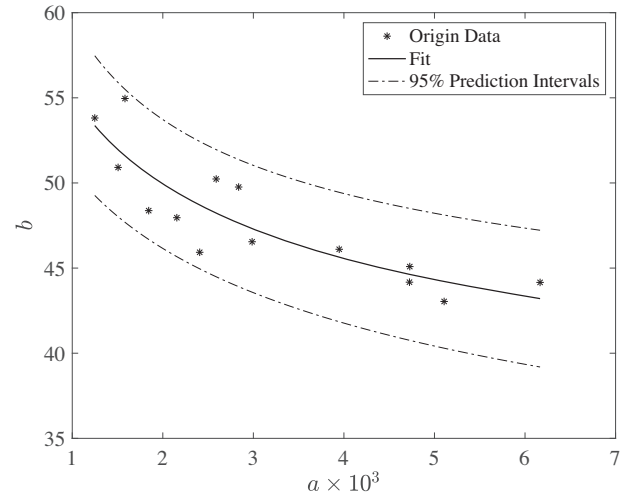


Figure 6. Scatter plots of parameters a and b and their fitting curves.

value of sunspots in each active cycle of 11–24, and the profile model parameters a , b , c for each cycle are acquired.

Fig. 6 shows the scatter plot of the profile parameters a and b corresponding to each cycle of sunspots and its fitting curve (black line) from the 11th to the 24th cycle. There is also a significant correlation between parameters a and b . We adopt the least-square method to approximate the regression expression between profile parameters a and b .

$$b = 22.4975 + 5.8032/a^{1/4}, \quad (11)$$

with a regression coefficient $R = 0.8789$, $F(1, n - 2)$ probability that the distribution is greater than the F value, $p = 3.4984e - 05 < \alpha$, thus the regression model is valid.

Section 3 provides an empirical equation (9) to predict the solar cycle by using the predicted peak value. For cycle 25, equation (9) becomes

$$RI_{\max} = 0.5125ab^3 + RI_{\min}(25).$$

Then, substitute (11) into (9) and $RI_{\min}(25) = 2.1271$, we get

$$RI_{\max} = 0.5125a(22.4975 + 5.8032/a^{1/4})^3 + 2.1271. \quad (12)$$

From the predicted value just obtained above, $RI_{\max}(25) = 140.1562$, we can get $a = 0.0023$ from equation (12), and then $b = 49.0460$ is solved from equation (11), so the forecast rising period for cycle 25 is

$$(t_{\max} - t_0)/12 = 4.3622 \text{ (yr)}. \quad (13)$$

Fig. 7 shows the predicted time sequence diagram of the sunspots number in cycle 25. It can be seen that the maximum of the 25th solar cycle will occur in March 2024, and its peak value $RI_{\max}(25)$ is 140.1563. This peak time is consistent with the result of Kitiashvili (2016) (2024 ± 1) but earlier than that of Petrovay et al. (2018), who suggested in 2024 August. The maximum of cycle 25 is 140.1563, larger than Upton & Hathaway (2018) predicted (110). Pesnell & Schatten (2018) used a polar precursor method to predict the solar activity in cycle 25, showing that maximum will reach 136 ± 48 .

6 CONCLUSION

In this paper, we propose a unimodal model to predict the complete sunspot cycle based on the comprehensive precursor information.

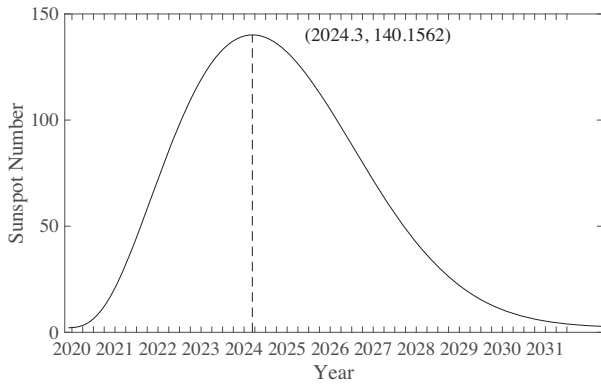


Figure 7. Prediction timing diagram of sunspot cycle 25. The dashed line gives the maximum and its occurrence time of sunspots in cycle 25.

The monthly average sunspot original data from February 1867 to October 2020 are processed by Gaussian smoothing to obtain the monthly smoothed sunspot number. The results are validated by the observed sunspots of cycle 24, and a long-term forecast for cycle 25 is made. The conclusion is as follows:

(1) Compared with the traditional 13-month moving average, the monthly smoothed data processed by Gaussian filter is closer to the overall trend of the raw data and the processed smoothed monthly mean value has less missing information, which is helpful to study the correlation between the parameters of the sunspot cycle.

(2) Through the permutation and combination of multiple parameters in precursor methods of solar cycle forecasting, the multiple regression technique is used to successfully achieve the periodical peak prediction. The regression coefficient (R) of the empirical model established in this paper is based on comprehensive precursor information and can reach 0.95. Compared with the traditional precursor method using the geomagnetic index, the new cycle peak prediction model significantly improves the prediction accuracy, adaptability, and stability.

(3) By adding a new parameter $RI_{\min}(n)$ to the original HWR function and using the predicted cycle peak, we provide a complete solar activity cycle profile showing the unimodal structure, and the fitting effect is significant. The peak value of cycle 25 predicted by this unimodal model will occur in 2024 March, with a peak of 140.2.

Note that the forecasting methods proposed in this paper are established on the basis of the valley (start value) of the known forecast solar cycle. Due to the limited amount of data, the proposed method needs more verification in the future.

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DATA AVAILABILITY

This research has made use of the NASA/IPAC sunspot number database which is open to download in the website <https://omniweb.gsfc.nasa.gov/>. The geomagnetic activity aa index data endorsed by IAGA can be freely downloaded from http://isgi.unistra.fr/data_download.php/.

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